RFC-HMS-2026-001 v2.1

Continuation: Missing Modules & Extended Specifications

50 Additional Modules: Clinical | Infrastructure | Patient Experience | Medical College

Complete ABAC Access Control | Equipment | Workflows | Regulatory Mapping

Parent: RFC-HMS-2026-001 v2.0 | February 2026 | Alagappa Group of Institutions

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PART A: Missing Hospital Clinical Departments & Units

A1. Mortuary & Post-Mortem Management

The mortuary is a critical hospital area with strict legal, ethical, and infection control requirements. No HMS in India covers this adequately.

A1.1 Functional Scope

Dead Body Receipt & Registration: Record time of death, declared by (doctor name + registration number), ward/location, cause of death (immediate, antecedent, underlying), manner (natural/accident/suicide/homicide/undetermined). Auto-generate death summary from IPD discharge summary module.

Cold Storage Management: Track body in cold storage (slot allocation, temperature monitoring). Maximum retention period: 72 hours (extendable by Medical Superintendent for MLC/unclaimed). Cold storage capacity tracking (typically 6-12 bodies for 450-bed hospital).

Body Release Workflow: Verify death certificate signed. For non-MLC: Release to family with identity verification (Aadhaar/ID match with registered attendant). For MLC: Police inquest completion required before release. Panchnama documentation. Body release register entry with photo documentation.

Post-Mortem Scheduling: Schedule autopsy with forensic medicine department. Track PM request (police/magistrate order), assigned forensic pathologist, date/time, PM report filing. Viscera preservation chain-of-custody with seal numbers and FSL dispatch tracking.

Unclaimed Body Protocol: 72-hour waiting period with police notification. Newspaper advertisement (track publication date). Municipal authority notification. Final disposal (cremation/burial) with photographic evidence. If medical college: Anatomy department allocation process (NMC rules on cadaver receipt).

Embalming Records: If applicable — embalming fluid used, procedure date, personnel, purpose (transport/preservation).

A1.2 ABAC Access Control

Role	Access	ABAC Conditions
Forensic Pathologist	V,C,U,A,P,X	FULL ACCESS to mortuary records. CREATE: PM reports, cause of death certificates. APPROVE: Body release for MLC cases. EXECUTE: Initiate PM, viscera dispatch. PRINT: PM reports (restricted distribution — only to requesting authority).
Duty Doctor (Death Declaration)	V,C	CREATE: Death declaration entry (time, cause). VIEW: Patient's terminal clinical records. CANNOT: Modify PM reports, release body without authorization. TIME: Must document within 1 hour of death.
Mortuary Attendant	V,C,X	VIEW: Body register, cold storage status. CREATE: Body receipt entry, storage slot assignment, cleaning/preparation records. EXECUTE: Body handover with supervisor oversight. CANNOT: Access clinical records or PM reports.
Medical Superintendent	V,A,P,E	APPROVE: Unclaimed body disposal, extended retention, body transport to another facility. EMERGENCY: Override body release in special circumstances (e.g., infectious disease requiring immediate cremation).
Police Officer (External)	V,P	VIEW: Death information, MLC register entry (NOT full clinical record). PRINT: Inquest report copy. ACCESS: Time-limited, escorted. ABAC: Only MLC cases, verified police ID.
Legal/MRD	V,P	VIEW: Death register, certificates. PRINT: Death certificates (for family, insurance). CANNOT: Modify clinical data.

A1.3 Equipment

Equipment	Qty	Cost (INR)	Calibration	PM Freq	Notes
Cold Storage Unit (Mortuary	2 (6-	5-15L each	6-Monthly	Monthly	Temperature 2-8°C monitoring with IoT

Refrigerator)	body each)				alert. Backup power mandatory.
Post-Mortem Table (Autopsy Table)	2	2-5L each	N/A	Monthly	Stainless steel, drainage, water supply, adjustable height
PM Instrument Set	4 sets	0.5-1L/set	N/A	6-Monthly	Dissection knives, bone saw, rib cutter, specimen jars
Autopsy Saw (Electric)	1	1-3L	Annual	6-Monthly	Oscillating, for skull opening
Weighing Scale (Organ)	1	0.1-0.3L	Annual	Annual	Capacity 0-5kg, calibrated for organ weights
Photography Equipment (Forensic)	1 set	0.5-2L	N/A	Annual	High-res camera, macro lens, scale ruler, evidence markers
Formaldehyde Dispenser/Storage	1	0.3-1L	N/A	Annual	Fume extraction mandatory, occupational safety
Body Transport Trolley	3	0.3-1L each	N/A	6-Monthly	Covered, easy-clean, elevator compatible
Exhaust/Ventilation System	1	2-5L	Annual	Quarterly	Negative pressure, HEPA filtration, odor control

Workflows:

WF-MORT-001: Death to Body Release (Non-MLC): Doctor declares death → Nursing documents → Body shifted to mortuary (transport log) → Cold storage assigned → Death certificate generated → Family notified → Identity verification → Billing clearance → Body release form signed → Handover with register entry.

WF-MORT-002: MLC Body Flow: Death declared → MLC flagged → Police intimated (auto-SMS + system notification) → Body to mortuary → Police inquest → If PM ordered: FM dept scheduled → PM conducted → PM report filed → Viscera dispatched to FSL (chain of custody) → Police clearance → Body release.

WF-MORT-003: Unclaimed Body: 72hr retention → Police FIR lodged → Newspaper notice → Municipal notice → 30-day wait → If unclaimed: Condemnation committee → Anatomy dept allocation OR cremation → Documentation with photos → Register closure.

A2. Catheterization Lab / Interventional Suite

The Cath Lab is a hybrid between OT and Radiology — it requires specific workflows not covered by either module.

A2.1 Functional Scope

Procedure Scheduling: Elective and emergency catheterization scheduling. Emergency STEMI pathway with door-to-balloon time tracking (target < 90 minutes). Slot management for multiple procedure types (diagnostic angiography, PCI, TAVI, pacemaker implant, EP study, peripheral angioplasty).

Procedure Documentation: Hemodynamic recording (pressures, saturations, gradients). Cine angiography storage (DICOM integration with PACS). Stent/device tracking (barcode scan of every device: stent, balloon, guidewire, pacemaker, ICD). Contrast volume tracking (cumulative per patient for nephrotoxicity risk). Radiation dose tracking (fluoroscopy time, DAP, air kerma — AERB compliance).

Device Inventory: High-value consumable tracking (stents INR 20K-3L each, pacemakers INR 50K-15L). Consignment stock management (vendor-owned devices stored in hospital, billed only on use). Expiry management (FEFO enforcement). Device implant registry (national/international reporting).

Post-Procedure Monitoring: Sheath removal timing, groin/radial site checks, hemostasis documentation, ambulation protocol, overnight monitoring for high-risk patients.

A2.2 ABAC Access Control

Role	Access	ABAC Conditions

Interventional Cardiologist	V,C,U,A,X,E	FULL ACCESS. CREATE: Procedure reports, device logs, hemodynamic records. APPROVE: Emergency procedure authorization. EXECUTE: Procedure documentation. EMERGENCY: Can activate STEMI pathway (auto-alerts: Cath Lab team, CCU, blood bank). Pre-approved break-glass for all cardiac emergencies.
Cath Lab Nurse	V,C,U,X	VIEW: Patient vitals, procedure schedule. CREATE: Nursing documentation, device usage log, specimen log. EXECUTE: Vital sign recording, medication administration during procedure (with physician order), hemostasis checks. CANNOT: Modify procedure reports.
Cath Lab Technician	V,C,X	VIEW: Equipment status, procedure schedule. CREATE: Equipment setup records, image acquisition logs, radiation dose records. EXECUTE: C-Arm/fluoroscopy operation (under physician direction), DICOM transfer. CANNOT: Access clinical notes beyond procedure indication.
Device Representative (Vendor)	V,C	VIEW: Consignment inventory for own company only. CREATE: Device delivery records. CANNOT: Access patient data, other vendor inventory, pricing. ACCESS: Time-limited, must be accompanied. Logged. NDA required.

A2.3 Equipment

Equipment	Qty	Cost (INR)	Calibration	PM Freq	Notes
Cath Lab System (Flat Panel/Biplane)	1	3-10 Cr	6-Monthly	Quarterly	AERB licensed. DICOM. Dose tracking. Hemodynamic integration.
Hemodynamic Recording System	1	10-30L	Annual	6-Monthly	Pressure transducers, oximetry, waveform recording
IVUS System (Intravascular Ultrasound)	1	15-40L	Annual	6-Monthly	Lesion assessment, stent optimization
FFR/iFR System	1	10-25L	Annual	6-Monthly	Functional assessment of coronary lesions
Intra-Aortic Balloon Pump	2	20-40L each	6-Monthly	Quarterly	Hemodynamic support during high-risk PCI
Temporary Pacemaker Generator	4	1-3L each	Annual	6-Monthly	Emergency pacing capability
Defibrillator (Cath Lab)	2	3-8L each	Annual	Quarterly	Biphasic, synchronized cardioversion
Contrast Injector (Power)	1	5-15L	Annual	6-Monthly	Programmable injection rates, pressure limits
Patient Monitor (Invasive)	2	5-15L each	Annual	Quarterly	IBP (multiple channels), SpO2, ECG, EtCO2
Lead Aprons + Thyroid Shields	10+	0.05-0.2L each	Annual	Annual	Integrity testing, weight optimization
Radiation Protection Screen	2	0.5-2L each	N/A	Annual	Ceiling/floor mounted, movable shields

A3. Endoscopy Suite

A3.1 Functional Scope

Procedure Management: Upper GI endoscopy (EGD), colonoscopy, sigmoidoscopy, ERCP, EUS (endoscopic ultrasound), bronchoscopy, cystoscopy. Scheduling with prep instructions auto-sent to patient (bowel prep for colonoscopy, fasting for EGD).

Scope Tracking & Reprocessing: Individual scope tracking by serial number. After each use: High-Level Disinfection (HLD) cycle logged. HLD parameters: Chemical used (OPA/glutaraldehyde/peracetic acid), concentration verification (MEC testing), soak time, temperature, rinse. Scope channel leak testing before reprocessing. Scope lifecycle tracking (number of uses, repairs, channel integrity). Mandatory culture surveillance (periodic scope tip cultures for infection control).

Sedation Documentation: Pre-sedation assessment (Mallampati, ASA class, fasting status). Sedation drugs administered (Midazolam, Propofol, Fentanyl — with doses and times). Continuous monitoring record (SpO2, HR, BP, ETCO2). Recovery scoring (Aldrete score). Discharge criteria met.

Biopsy & Specimen Tracking: Biopsy site documentation (diagram/photo), specimen labeling with barcode, chain of custody to pathology, histopathology result linking back to endoscopy report.

A3.2 ABAC Access Control

Role	Access	ABAC Conditions
Endoscopist (GI Surgeon/Physician)	V,C,U,A,X,P	FULL ACCESS. CREATE: Procedure reports with images, biopsy records. APPROVE: Procedure scheduling. EXECUTE: Complete procedure documentation. PRINT: Reports with endoscopic images. VIEW: Patient history, medications (anticoagulants critical), prior endoscopies.
Endoscopy Nurse	V,C,U,X	CREATE: Pre-procedure checklist, sedation monitoring, recovery documentation, scope reprocessing logs. EXECUTE: Medication administration (sedation, under physician order), biopsy specimen handling. CANNOT: Finalize procedure reports.
Scope Reprocessing Technician	V,C,X	VIEW: Scope ID, reprocessing schedule. CREATE: HLD cycle records, leak test results, MEC test results, culture results. EXECUTE: Release scope for use (ONLY if all reprocessing criteria met — system enforces). CANNOT: Access patient data at all. Equipment-only access.
Infection Control Nurse	V,P	VIEW: Scope reprocessing records, culture surveillance results, HLD compliance metrics. PRINT: Infection control audit reports. CANNOT: Modify reprocessing records.

A3.3 Equipment

Equipment	Qty	Cost (INR)	Calibration	PM Freq	Notes
Video Endoscopy Tower (Processor + Source + Monitor)	2	15-40L each	Annual	6-Monthly	HD/4K, recording capability, DICOM image capture
Gastroscope (Diagnostic)	4	5-12L each	Annual	After each use (HLD)	Leak test before every reprocessing cycle
Colonoscope	4	6-15L each	Annual	After each use (HLD)	Variable stiffness, wide angle
Duodenoscope (ERCP)	2	8-20L each	Annual	After each use (HLD)	Elevator mechanism — highest reprocessing risk
EUS Scope (Linear/Radial)	1	15-30L	Annual	After each use (HLD)	Ultrasound integration
Bronchoscope (Flexible)	2	5-12L each	Annual	After each use (HLD)	Diagnostic + therapeutic
AER (Automated Endoscope Reprocessor)	2	5-15L each	6-Monthly	Monthly	Automated HLD cycle with documentation. Must log every cycle.
Endoscopy Patient Monitor	2	2-5L each	Annual	Quarterly	SpO2, ETCO2, ECG, NIBP for sedation
Electrosurgical Generator (Endoscopic)	1	3-8L	Annual	6-Monthly	Polypectomy, hemostasis, APC
Insufflator (CO2)	2	2-5L each	Annual	6-Monthly	CO2 preferred over air for patient comfort

A4. Day Care / Short Stay Unit

A4.1 Functional Scope

Admission Types: Chemotherapy (day care), minor surgical procedures (excision biopsy, hernia under LA, cataract), blood transfusion, IV antibiotic infusion, observation (post-procedure, post-fall), dialysis day care, endoscopy recovery, pain management procedures.

Different from IPD: No overnight stay (< 24 hours). Simplified admission process. Pre-scheduled procedures. Different billing structure (package-based). Separate bed management (recliner chairs for some procedures). Rapid turnaround (clean, prep, use, clean cycle).

Key Workflows: Pre-procedure assessment (same-day or prior visit). Day care admission (simplified form, pre-approved by scheduling). Procedure/treatment. Post-procedure monitoring (recovery area with nurse observation). Discharge criteria assessment. Same-day discharge with take-home instructions. Follow-up scheduling.

A4.2 ABAC Access Control

Role	Access	ABAC Conditions
Day Care Coordinator	V,C,U,A,X	VIEW: Day care schedule, bed availability, patient list. CREATE: Day care admission records. UPDATE: Scheduling changes. APPROVE: Priority overrides, discharge clearance. EXECUTE: Bed allocation, patient flow management.
Treating Doctor	V,C,U,A,X,P	All clinical access for day care patients under their care. CREATE: Procedure notes, post-procedure orders, discharge orders. APPROVE: Discharge fitness certification.
Day Care Nurse	V,C,U,X	VIEW: Assigned patients, vitals, medication orders. CREATE: Vitals, I/O, nursing observation, medication administration, pre-procedure checklist. EXECUTE: Chemotherapy administration (specialized certification required — ABAC: user.chemo_certification = active for chemo patients).
Patient	V,P	VIEW: Appointment details, preparation instructions, estimated cost. PRINT: Appointment confirmation, preparation checklist.

A5. Palliative Care & End-of-Life Management

A5.1 Functional Scope

Patient Enrollment: Palliative care referral workflow. Prognosis documentation. Goals-of-care discussion documentation (who was present, what was discussed, patient/family wishes). Advance Directive recording (legally binding under Indian law if properly documented).

DNR/DNAR Orders: Do Not Resuscitate order workflow. Requires: Consultant documentation of medical futility, patient/surrogate consent, witnessed discussion. DNR order clearly visible on patient dashboard (cannot be missed during emergency). Regular review (every 48 hours for IPD, every visit for OPD). DNR can be revoked by patient at any time (instant system update).

Pain Management: WHO Pain Ladder tracking (Step 1: Non-opioid → Step 2: Weak opioid → Step 3: Strong opioid). Pain assessment scores (NRS/VAS) tracking over time. Opioid dose escalation documentation. NDPS Act compliance for morphine/fentanyl prescribing. Opioid consumption register (ward-wise, patient-wise).

Home-Based Palliative Care: Home visit scheduling, visit documentation (mobile app), home medication management, telemedicine follow-up, caregiver training records, bereavement support tracking.

Organ Donation Integration: When patient is potential organ donor (brain death criteria being met), auto-notify ROTTO/SOTTO (Regional/State Organ Transplant Organisation). THOA (Transplantation of Human Organs Act) compliance. Donor registry check. Family counseling documentation.

A5.2 ABAC Access Control

Role	Access	ABAC Conditions
Palliative Care Physician	V,C,U,A,X,E	FULL clinical access. CREATE: Palliative care plan, DNR orders (requires consultant designation), advance directive documentation. APPROVE: Opioid escalation beyond standard threshold. EXECUTE: ROTTO notification for organ donation. SPECIAL: Can prescribe high-dose opioids (NDPS compliance). Break-glass for cross-department terminal patient access.
Palliative Nurse	V,C,U,X	VIEW: Patient's palliative plan, medication orders. CREATE: Pain assessments, symptom diary, home visit records, caregiver education records. EXECUTE: Medication administration. CANNOT: Initiate DNR, modify opioid prescriptions.
Social Worker	V,C	VIEW: Patient demographics, insurance, social assessment. CREATE: Financial counseling, government scheme applications, bereavement support records. CANNOT: Access clinical details beyond diagnosis summary.
Transplant Coordinator	V,C,X	VIEW: Potential donor patient details (ONLY when brain death declaration initiated). CREATE: Donor evaluation records, family consent documentation, ROTTO communication. EXECUTE: Initiate organ allocation process. ABAC: Access ONLY available when patient.brain_death_protocol = active. No access to general patient records.

A6. Nuclear Medicine (If Applicable)

A6.1 Functional Scope

Radiopharmaceutical Management: Source procurement (Tc-99m generator, I-131, F-18 FDG for PET). Dose calibration and dispensing. Patient dose administration records. Radioactive waste management (decay storage, half-life tracking). AERB source inventory and movement register.

Patient Safety: Pregnancy verification before nuclear medicine study. Breastfeeding cessation advisory. Post-therapy isolation (I-131 therapy — patient must remain isolated until radiation level < threshold). Discharge radiation survey. Public exposure risk documentation.

Imaging: Gamma camera, SPECT, PET-CT acquisition. DICOM integration with PACS. Quantitative analysis (SUV values for PET). Hybrid imaging protocols.

A6.2 ABAC Access Control

Role	Access	ABAC Conditions
Nuclear Medicine Physician	V,C,U,A,X,P	FULL ACCESS. CREATE: Study reports, therapy plans, source requisitions. APPROVE: Patient discharge after therapy (radiation survey clearance). EXECUTE: Dose administration authorization. SPECIAL: AERB authorized user.
Nuclear Medicine Technologist	V,C,X	CREATE: Dose preparation records, acquisition parameters, patient injection records, waste logs. EXECUTE: Image acquisition, dose dispensing (under physician supervision). CANNOT: Report, approve discharge. AERB: Must have valid radiation worker certification.
RSO (Radiation Safety Officer)	V,C,U,A,P	VIEW: All radiation-related records, source inventory, personnel dosimetry. CREATE: Area monitoring records, incident reports. APPROVE: Source procurement, waste disposal. PRINT: AERB compliance reports. CANNOT: Access clinical patient data.

A7. Physical Medicine & Rehabilitation (PMR)

A7.1 Functional Scope

Multi-disciplinary Rehabilitation: Physiotherapy (already partially covered — expanded here), Occupational Therapy (ADL training, upper limb rehabilitation, cognitive rehabilitation, assistive device assessment), Speech & Language Therapy (dysarthria, aphasia, dysphagia assessment — FEES/VFSS, voice therapy, stuttering management, AAC device trials), Prosthetics & Orthotics (prosthesis prescription, fitting, follow-up, gait analysis).

Disability Assessment: Disability certificate generation (as per RPWD Act 2016). Assessment by certified disability board. Percentage calculation per body system. Certificate format per government mandate. Certificate validity tracking (temporary vs permanent).

Outcome Measurement: Standardized scales: FIM (Functional Independence Measure), Barthel Index, Modified Rankin Scale, Berg Balance Scale, 6-Minute Walk Test, Glasgow Outcome Scale. Pre/post treatment comparison charts. Goal attainment scaling. Discharge planning with community integration plan.

A7.2 ABAC Access Control

Role	Access	ABAC Conditions
PMR Specialist	V,C,U,A,X,P	FULL ACCESS for rehabilitation patients. CREATE: Rehabilitation plans, disability assessments, orthotic/prosthetic prescriptions. APPROVE: Disability certificates (only certified board members). PRINT: Disability certificates, progress reports.
Physiotherapist	V,C,U,X	VIEW: Patient diagnosis, relevant imaging, surgical notes (for post-surgical rehab). CREATE: PT assessment, treatment sessions, outcome scores. UPDATE: Treatment plan modifications. CANNOT: Prescribe medications, order investigations, view unrelated clinical data.
Occupational Therapist	V,C,U,X	VIEW: Patient functional status, cognitive assessments. CREATE: OT assessment, ADL training records, assistive device trials. CANNOT: Access detailed surgical/medical notes beyond functional relevant data.
Speech Therapist	V,C,U,X	VIEW: Patient diagnosis, neurology notes, ENT examination. CREATE: Speech assessment, swallowing assessment, therapy session records. CANNOT: Access data beyond relevant specialty.
Prosthetics/Orthotics Technician	V,C,X	VIEW: Prescription details, measurement records. CREATE: Device fabrication records, fitting notes, modification records. EXECUTE: Device delivery documentation. CANNOT: Access clinical records.

A8. Audiology & Speech-Language Pathology

A8.1 Functional Scope

Diagnostic Audiology: Pure Tone Audiometry (PTA), speech audiometry, impedance audiometry (tympanometry + acoustic reflexes), BERA (Brainstem Evoked Response Audiometry), OAE (Otoacoustic Emissions), ASSR. Newborn Hearing Screening Program (NHSP) — universal newborn screening with OAE, referral tracking for failed screens.

Hearing Aid Services: Hearing aid trial, fitting, programming (with audiogram integration), Real Ear Measurement (REM), follow-up adjustments, user satisfaction tracking. Government scheme integration (ADIP scheme for hearing aids).

Cochlear Implant Program: Candidacy assessment, pre-operative evaluation, device selection (consignment/procurement), intra-operative monitoring, switch-on, mapping sessions, rehabilitation tracking.

A8.2 ABAC Access Control

Role	Access	ABAC Conditions
Audiologist	V,C,U,A,P,X	FULL ACCESS to audiology records. CREATE: Audiometric test results, hearing aid prescriptions, cochlear implant evaluations. APPROVE: NHSP screening results (pass/refer). PRINT: Audiograms, reports. EXECUTE: Release test results to ENT department.
Speech-Language Pathologist	V,C,U	VIEW: Patient diagnosis, audiology results. CREATE: Speech-language assessments, therapy plans, session records. CANNOT: Modify audiometric results. Shared department with audiology.
ENT Surgeon	V,P	VIEW: Audiometric results for own patients. PRINT: Reports for surgical planning. CANNOT: Modify audiology records.
NHSP Screener (Trained Nurse)	V,C,X	VIEW: Newborn list, screening schedule. CREATE: OAE screening results (pass/refer). EXECUTE: Schedule BERA referral for failed screens. CANNOT: Interpret results or access other audiology data.

A8.3 Equipment

Equipment	Qty	Cost (INR)	Calibration	PM Freq	Notes
Pure Tone Audiometer	3	2-6L each	Annual	Annual	Air + bone conduction, masking, speech
Impedance Audiometer (Tympanometer)	2	2-5L each	Annual	Annual	Tympanometry, acoustic reflexes, ETF
BERA/ABR System	1	5-15L	Annual	6-Monthly	Click + tone-burst, threshold estimation
OAE System (TEOAE + DPOAE)	2	2-5L each	Annual	Annual	Screening (NHSP) + diagnostic
ASSR System	1	5-10L	Annual	6-Monthly	Frequency-specific threshold for pediatric
Sound-Treated Room (Audiometric Booth)	2	5-15L each	Annual ambient check	Annual	ISO 8253 ambient noise standards
Hearing Aid Programmer + REM	1 set	3-8L	Annual	Annual	NOAH compatible, all manufacturer software
Cochlear Implant Programming System	1	5-10L	Annual	6-Monthly	Multi-manufacturer: Cochlear, MED-EL, AB
Video Otoscope	2	0.5-2L each	N/A	Annual	Documentation, patient education, telemedicine

A9. Psychiatry — Mental Healthcare Act 2017 Compliance

A9.1 Functional Scope (Beyond Standard OPD/IPD)

Mental Healthcare Act (MHCA) 2017 Compliance: This is a complex law that fundamentally changes how psychiatric patients must be managed in the system.

Advance Directives: Patient can create an Advance Directive specifying how they want to be treated during future mental health crises. System must: Store advance directive, verify it's registered with Mental Health Board, display prominently when patient presents, allow revocation anytime, track nominated representative details.

Nominated Representative: Every mental illness patient can nominate a representative. System tracks: Who is nominated, contact details, relationship, notification preferences. NR must be involved in major treatment decisions. NR can request review of treatment by Mental Health Review Board.

Admission Categories (MHCA): Independent Admission (patient's own request, no restriction on discharge). Supported Admission (with nominated representative support, up to 30 days initially, extendable). System must track admission type, review dates, mandatory assessments, and auto-generate notifications to Mental Health Board for supported admissions > 30 days.

ECT (Electroconvulsive Therapy) Register: MHCA allows ECT only with muscle relaxant and anesthesia (modified ECT). System must: Record consent (informed consent with capacity assessment), document indication, track sessions (number, laterality, stimulus parameters), monitor response (HAM-D, BPRS scores), report to Mental Health Board.

Substance Abuse Management: De-addiction programs, withdrawal protocols (CIWA-Ar for alcohol, COWS for opioids), medication-assisted treatment (Buprenorphine, Naltrexone, Disulfiram), relapse tracking, NDPS Act compliance for opioid substitution.

Seclusion & Restraint: MHCA has strict rules. Physical restraint only as last resort. Must be ordered by psychiatrist. Maximum 4 hours without review. Continuous monitoring. Detailed documentation of reason, duration, patient condition during restraint. Monthly reporting to Mental Health Board.

A9.2 ABAC Access Control

Role	Access	ABAC Conditions
Psychiatrist	V,C,U,A,X,E	FULL ACCESS to psychiatric records. SPECIAL: Psychiatric records are ISOLATED from main hospital record. Only psychiatry department staff can view. CREATE: MHCA-compliant admission forms, treatment plans, ECT orders, advance directive verification. APPROVE: Supported admission, restraint orders, involuntary treatment. EXECUTE: Mental Health Board notifications.
Clinical Psychologist	V,C,U	VIEW: Psychiatric records, treatment plans. CREATE: Psychometric assessments (IQ, personality, neuropsychological), therapy notes (CBT, DBT session records), psychological reports. CANNOT: Prescribe, order investigations, approve admissions.
Psychiatric Nurse	V,C,U,X	VIEW: Patient care plans, medication orders. CREATE: Nursing observations (mental status monitoring, risk assessment — suicide/violence, daily living assessment), medication administration, restraint monitoring logs (every 30 min). EXECUTE: Administer medication, emergency PRN medication.
Social Worker (Psychiatry)	V,C	VIEW: Social assessment, family dynamics, occupational history. CREATE: Social evaluation, rehabilitation planning, vocational assessment, family therapy notes. CANNOT: Access detailed clinical psychiatric notes.
Nominated Representative	V	VIEW: Treatment plan summary, admission details, rights under MHCA. CANNOT: Access detailed clinical notes. CHANNEL: Portal access with OTP verification. SPECIAL: Can file review petition (system generates MHRB format).
Mental Health Review Board (External)	V,P	VIEW: Specific case documents as per Board request. PRINT: Reports in MHRB format. ACCESS: Case-specific only, time-limited, provisioned by Medical Superintendent. Full audit.

Critical ABAC Rules for Psychiatry:

Rule PSY-001: COMPLETE DATA ISOLATION. Psychiatric records stored in separate encrypted partition. Even Hospital Administrator / Medical Superintendent cannot view without break-glass. ER doctors during psychiatric emergency get limited view (current medications, allergies, suicide risk level — not detailed therapy notes).

Rule PSY-002: Substance abuse records: Additional isolation layer. Even within psychiatry, substance abuse details visible only to treating team. Insurance queries about substance abuse must be handled per MHCA Section 23 (no discrimination).

Rule PSY-003: ECT register: Separate access control. Only treating psychiatrist + anesthesiologist can view. ECT count per patient visible to Mental Health Board auditor.

A10. Clinical Psychology Unit

A10.1 Functional Scope

Psychometric Testing: Intelligence testing (WAIS, WISC, Binet-Kamat), personality assessment (MMPI-2, 16PF, Rorschach, TAT), neuropsychological battery (NIMHANS battery, ACE-III, MoCA, Trail Making, Wisconsin Card Sorting), memory assessment (PGI Memory Scale, WMS), behavioral assessment, developmental assessment (VSMS, BSID, CARS for autism). Test material security — controlled access, no copying, restricted printing.

Therapy Documentation: Individual therapy sessions (CBT, DBT, EMDR, psychodynamic), group therapy sessions, couple/family therapy, outcome tracking (PHQ-9, GAD-7, BDI pre/post), session video recording (with consent, for supervision).

Fitness Assessments: Pre-employment psychological fitness, fitness to stand trial (forensic psychology), child custody evaluation, learning disability assessment, RPWD Act disability certification support.

A10.2 ABAC Access Control

Role	Access	ABAC Conditions
Clinical Psychologist	V,C,U,A,P	FULL ACCESS to psychology records. CREATE: Assessment reports, therapy notes, psychometric test results. APPROVE: Assessment report finalization. PRINT: Reports (with appropriate disclaimers about test security). SPECIAL: Raw test data (scoring sheets, protocols) NEVER shared — only interpreted report shared with referring clinician.
Psychology Trainee	V,C	VIEW: Assigned cases only. CREATE: Session notes (marked as trainee, requires supervisor co-sign), test administration records. CANNOT: Finalize reports, access raw test materials independently.
Referring Clinician	V,P	VIEW: Final assessment report and recommendations ONLY. Cannot view raw test data, detailed therapy process notes, or session recordings. PRINT: Report summary. ABAC: data.type = psych_final_report only.

PART B: Hospital Infrastructure Management Modules

B1. Medical Gas Pipeline System (MGPS) Management

B1.1 Functional Scope

This became the single most critical infrastructure post-COVID. Hospitals lost patients due to oxygen supply failures. Our system must prevent this.

Oxygen Plant Monitoring: PSA (Pressure Swing Adsorption) plant real-time monitoring — O2 purity (>93%), pressure, flow rate, operating temperature. Liquid Medical Oxygen (LMO) tank level monitoring. Cylinder manifold status (online/standby/empty). Auto-switchover monitoring (manifold A to B). Daily O2 consumption per department (calculated from flow meters).

Pipeline Monitoring: Zone valve status per floor/department. Pipeline pressure at key points (outlet pressure 3.5-4 bar). Leak detection alerts. Alarm panel integration (area alarm, master alarm). Gas outlet point inventory per bed/OT/ICU.

Other Gases: Nitrous Oxide (N2O) for OT — cylinder tracking, consumption. Medical Air — compressor status, dew point, oil-free certification. Suction/Vacuum — pump status, pressure at outlets. CO2 — for laparoscopy.

Compliance: Drug license for medical gases (O2, N2O are drugs under D&C Act). PESO (Petroleum and Explosives Safety Organisation) compliance for LMO storage. Indian Pharmacopoeia standards for gas purity. Daily log maintenance (IP requirement).

B1.2 ABAC Access Control

Role	Access	ABAC Conditions
MGPS Technician	V,C,U,X	VIEW: All gas system parameters, alarms, cylinder inventory. CREATE: Daily logs, maintenance records, alarm response records. UPDATE: Cylinder manifold changeover, refill status. EXECUTE: Emergency manifold switchover, alarm acknowledgment. 24/7 access.
MGPS Supervisor	V,C,U,A,P	All technician permissions + APPROVE: Maintenance shutdown scheduling, vendor service reports. PRINT: Compliance reports, consumption reports. CREATE: Incident reports for gas supply failures.
Medical Superintendent	V,A,P,E	VIEW: O2 consumption dashboard, reserve status. APPROVE: Emergency O2 procurement. EMERGENCY: Override normal supply protocols during crisis. PRINT: Regulatory compliance reports.
ICU/OT Nurse	V	VIEW: Gas supply status for their area (green/yellow/red indicator on bedside display). ALERT: Auto-notified if zone pressure drops. CANNOT: Access technical gas system data.
Facilities Manager	V,C,U,A,P	VIEW: All infrastructure gas systems. CREATE: Preventive maintenance schedules. APPROVE: Annual maintenance contracts for gas systems. PRINT: Facility management reports including gas systems.

B1.3 Equipment

Equipment	Qty	Cost (INR)	Calibration	PM Freq	Notes
PSA Oxygen Plant	1	30-80L	6-Monthly	Monthly	O2 purity analyzer: daily check. Auto-dump if purity < 93%
LMO Tank + Vaporizer	1	15-30L	Annual	Monthly	PESO compliance. Level sensor with IoT alert.
Cylinder Manifold (O2/N2O/Air)	3 sets	3-8L/set	Annual	6-Monthly	Auto-changeover, pressure sensors
Medical Air Compressor (Oil-	2	5-15L each	Annual	Monthly	Dew point monitoring, redundant

Free)	(1+1)				operation
Vacuum Plant	2 (1+1)	5-12L each	Annual	Monthly	Suction pressure at outlets: 400-500 mmHg
Pipeline Pressure Sensors (IoT)	50+	0.05-0.2L each	6-Monthly	Annual	At every zone valve + critical outlets. Real-time dashboard.
Gas Purity Analyzer	1	3-8L	Monthly	Monthly	Continuous O2 purity monitoring at plant output
Alarm Panel (Area + Master)	Multiple	1-3L total	Annual	6-Monthly	Visual + audible alarms per zone

B2. Biomedical Waste Management (BMW Rules 2016)

B2.1 Functional Scope

This is a LEGAL requirement with criminal penalties for non-compliance (up to 5 years imprisonment under Environment Protection Act).

Color-Coded Segregation Tracking: Yellow bag (human anatomical waste, soiled waste, chemical waste, chemotherapy waste, discarded medicines). Red bag (contaminated recyclable waste — tubing, bottles, syringes without needles). White container (sharps — needles, scalpels, broken glass). Blue box (glassware, metallic implants). System tracks weight of each category per department per day.

Manifest System: Every consignment to authorized waste handler must have manifest (like a challan). System generates: Date, quantity by category, vehicle number, transporter name, treatment facility. Manifest signed by hospital + transporter. Return manifest from treatment facility tracked (proof of proper disposal).

Sharp Container Management: Track each sharp container from deployment to disposal. Fill level monitoring (3/4 full = replace). Needle stick injury reporting linked to specific container/ward.

Regulatory Reporting: SPCB (State Pollution Control Board) quarterly returns (Form IV). Annual report to SPCB. HCF (Healthcare Facility) registration. CPCB reporting for common bio-medical waste treatment facility.

Mercury Management: Mercury spill protocol. Mercury-containing equipment inventory (thermometers, BP apparatus). Phased replacement tracking (as per UNEP Minamata Convention compliance).

B2.2 ABAC Access Control

Role	Access	ABAC Conditions
BMW Coordinator (Infection Control)	V,C,U,A,P,X	FULL ACCESS. CREATE: Daily waste generation records, manifest, incident reports. APPROVE: Waste transporter verification, disposal records. EXECUTE: SPCB report generation. PRINT: All BMW reports. VIEW: Department-wise generation analytics.
Ward Nurse/Housekeeping Supervisor	V,C	VIEW: Ward-specific waste generation. CREATE: Daily segregation compliance records, sharp container replacement requests. CANNOT: Access hospital-wide waste data or modify manifests.
Housekeeping Staff	C,X	CREATE: Waste collection records (time, weight per bag, ward). EXECUTE: Waste transport to central storage. CANNOT: View any clinical data. Mobile app for scanning waste bag barcodes.
SPCB Inspector (External)	V,P	VIEW: BMW records, manifests, authorization certificates. PRINT: Compliance reports. ACCESS: Time-limited, read-only. Provisioned by Hospital Administration.

B3. Fire Safety & Emergency Preparedness

B3.1 Functional Scope

Fire Equipment Inventory: Fire extinguishers (type, location, expiry, last refill, next refill), fire hydrants, hose reels, sprinkler system zones, smoke detectors, fire alarm pull stations, fire doors, emergency lights, exit signs. Each item tagged with QR code for inspection scanning.

Mock Drill Management: NABH requires quarterly fire drills. System tracks: Drill date, participating departments, evacuation time, issues identified, corrective actions, attendance. Generates mock drill report in NABH format.

Fire NOC Management: Fire NOC validity tracking, renewal alerts, compliance requirements per local fire authority.

Emergency Response Protocol: Code Red activation workflow. Zone-wise evacuation plan (digital display at nurse stations). Patient categorization during evacuation (ambulatory/wheelchair/stretcher/critical). RACE protocol documentation (Rescue, Alarm, Contain, Extinguish/Evacuate).

B3.2 ABAC Access Control

Role	Access	ABAC Conditions
Fire Safety Officer	V,C,U,A,P,X	FULL ACCESS to fire safety module. CREATE: Inspection records, drill reports, training records. APPROVE: Equipment maintenance completion, NOC renewal submission. EXECUTE: Code Red activation (triggers hospital-wide alerts). PRINT: All fire safety reports.
Facilities/Maintenance Staff	V,C,U,X	VIEW: Fire equipment inventory, maintenance schedule. CREATE: Maintenance/refill records, inspection checklists. EXECUTE: Equipment servicing confirmation. CANNOT: Approve or activate emergency protocols.
Ward In-Charge (Nurse/Doctor)	V,C	VIEW: Fire equipment locations in their area, evacuation plan. CREATE: Ward-level drill participation records. Accessed via: Fire safety page on ward dashboard.
All Staff	V	VIEW: Evacuation routes, assembly points, fire warden contacts. MANDATORY: Annual fire safety training completion tracked in HR module.

B4. Water Quality & Energy Management

B4.1 Water Management

Water Quality Testing: Bacteriological (coliform count), chemical (chlorine residual, pH, TDS, hardness). Testing frequency: Weekly for patient care areas, daily for dialysis unit, monthly for general areas. Sample location tracking. Result trending. Exceedance alerts.

Dialysis Water (Special): RO plant monitoring — pre-treatment (softener, carbon filter), RO membrane performance (rejection rate), post-treatment (UV, deionization). Water quality parameters: Conductivity < 10 µS/cm, endotoxin < 0.25 EU/mL, bacteria < 100 CFU/mL. Continuous monitoring with IoT sensors. Alert if parameter out of range — dialysis must stop.

Legionella Prevention: Hot water system temperature monitoring (> 60°C at source, > 50°C at outlets). Water tank cleaning schedule (6-monthly per NABH). Dead-end pipe flushing schedule. Temperature records for NABH audit.

Consumption Monitoring: Department-wise water consumption. Leak detection. Rainwater harvesting monitoring (if applicable). STP (Sewage Treatment Plant) treated water quality and reuse tracking.

B4.2 Energy Management

Power Source Monitoring: Grid power status, DG set (runtime, fuel consumption, load percentage, auto-start testing), UPS systems per critical area (battery health, runtime on battery, transfer time), solar panel output (if installed). Power factor monitoring for electricity board compliance.

Critical Area Power Mapping: OT, ICU, NICU, Blood Bank, Lab (freezers), Server Room — each must have dual power source with < 15 second switchover. System tracks: Last switchover test, backup runtime at current load, fuel reserve (hours).

CEA Compliance: Central Electricity Authority safety regulations. Electrical safety testing schedule (ELCB testing, earthing verification, insulation resistance). Electrical accident/incident reporting.

B4.3 ABAC Access Control

Role	Access	ABAC Conditions
Facilities Manager	V,C,U,A,P,X	FULL ACCESS to all infrastructure modules. CREATE: Maintenance schedules, testing records, compliance reports. APPROVE: Vendor work completion, procurement. EXECUTE: Emergency protocols (water supply shutdown, DG activation). PRINT: All facility reports for NABH/NMC.
Electrical/Plumbing Technician	V,C,U,X	VIEW: Assigned work orders, equipment status. CREATE: Work completion records, testing results, meter readings. EXECUTE: Equipment operation, maintenance tasks. CANNOT: Approve expenditure or access clinical data.
Dialysis Unit In-Charge	V	VIEW: RO water quality dashboard (real-time). ALERT: Auto-notified if water quality parameters out of range. ACTION: Can halt dialysis procedure if water quality alert. CANNOT: Modify water system records.
Infection Control Officer	V,P	VIEW: Water quality test results, Legionella monitoring, cooling tower maintenance. PRINT: Infection control audit reports. CANNOT: Modify facility records.

B5. Laundry Management

B5.1 Functional Scope

Linen Lifecycle: Track every linen item (bed sheet, pillow cover, patient gown, OT linen, surgical drapes) from procurement to condemnation. Barcode/RFID tracking. Expected lifecycle: 100-150 washes for bed linen, 50-75 for OT linen. Auto-replacement alert when lifecycle reached.

Collection & Sorting: Ward-wise soiled linen collection (weight-based). Infection classification: Regular soiled, blood/body fluid contaminated, isolation linen (separate handling). Linen audit: What went to laundry vs what came back (loss tracking).

Wash Process: Barrier washing for infected linen (thermal + chemical disinfection). Wash formula per category (temperature, detergent, cycle time). Water quality monitoring for laundry. Microbiological monitoring of clean linen (periodic cultures).

Distribution: Clean linen par levels per ward. Demand forecasting based on occupancy. Emergency linen (for blood/fluid spills). OT linen pack management (pack composition, sterilization if required).

B5.2 ABAC Access Control

Role	Access	ABAC Conditions
Laundry Manager	V,C,U,A,P	FULL ACCESS. CREATE: Wash records, distribution records, condemnation recommendations. APPROVE: Linen procurement, vendor laundry quality. PRINT: Linen audit reports. VIEW: All ward linen status.
Laundry Staff	V,C,X	VIEW: Collection schedule, wash formulas. CREATE: Collection weight records, wash cycle records. EXECUTE: Linen release after quality check. CANNOT: Access patient data.
Ward Nurse	V,C	VIEW: Linen availability for their ward. CREATE: Linen request (extra/emergency), soiled linen handover record. CANNOT: Access laundry operational data.
Infection Control	V,A,P	VIEW: Wash parameters, microbiological results, infected linen handling records. APPROVE: Wash protocol changes. PRINT: Infection control audit. CANNOT: Modify laundry operations.

B6. Security & Access Control Integration

B6.1 Functional Scope

Physical Access Control: Integration with door access system (card/biometric/PIN). Zone definitions: Public (reception, waiting areas), Semi-restricted (wards, OPD), Restricted (ICU, OT, pharmacy, blood bank, server room), Highly Restricted (narcotic store, radiation area, medical records). Access log for each zone.

CCTV Integration: Camera inventory management (location, type, recording status). Incident-linked video retrieval (when an incident is reported, system auto-tags relevant camera timestamps). Recording retention: 30 days general, 90 days high-security, permanent for incidents.

Infant Security (NICU/Nursery): Infant tagging system (RFID wristband on baby and mother). Zone alert if infant tag exits designated area. Mother-baby matching verification before discharge. Missing infant protocol (Code Pink activation).

Patient Security: Absconder patient protocol (psychiatric, MLC patients). Elopement risk assessment integration from nursing module. Wander guard for dementia patients. Visitor management for restricted patients (MLC, VIP).

Threat Response: Code Silver (weapon/hostage), Code Orange (hazmat), Code Black (bomb threat). Protocol activation workflows, communication tree, all-clear procedures.

B6.2 ABAC Access Control

Role	Access	ABAC Conditions
Security Head	V,C,U,A,P,X	FULL ACCESS to security module. CREATE: Incident reports, access policy changes. APPROVE: Access card provisioning, CCTV review requests. EXECUTE: Code activation (Code Pink/Silver/Orange/Black). PRINT: Security reports. VIEW: All access logs, CCTV status.
Security Guard	V,C,X	VIEW: Visitor log, access alerts. CREATE: Visitor entry records, patrol records, incident observations. EXECUTE: Gate entry/exit logging, code response actions. CANNOT: Access clinical data, CCTV playback, or modify access policies.
HR (Access Card Management)	V,C,U,X	VIEW: Employee access card status. CREATE: New access cards for new employees. UPDATE: Access zone assignments per role. EXECUTE: Card activation/deactivation. CANNOT: View security incidents or CCTV.
NICU Nurse	V,X	VIEW: Infant tag status for NICU babies. EXECUTE: Mother-baby verification scan during feeding/discharge. ALERT: Auto-receives Code Pink alert if infant tag triggers zone breach.
IT Admin	V,U	VIEW: System health of access control, CCTV storage. UPDATE: System configuration, camera assignments. CANNOT: View CCTV footage (security authorization required for playback). Technical access only.

PART C: Patient Experience & Operational Modules

C1. Visitor Management

C1.1 Functional Scope

Visitor Registration: Photo capture, ID verification (Aadhaar/voter ID), relationship to patient, contact number. Visitor pass generation (time-limited, ward-specific). QR code on visitor pass for entry/exit scanning.

Visiting Hours Enforcement: Configurable visiting hours per ward type (General: 4-6 PM, ICU: 10 AM + 4 PM for 15 min, NICU: per protocol). System auto-rejects visitor entry outside hours (unless authorized override). Outbreak mode: Complete visitor restriction with exceptions only by treating doctor.

Visitor Count Control: Maximum visitors per patient per slot (configurable: typically 2). Visitor swap (one exits, another enters). Infection control: Hand hygiene compliance before entry (integrated with hand hygiene monitoring if available).

Special Visitor Categories: Religious/spiritual visitor, legal counsel (MLC patients), government official, press (restricted), VIP visitor (separate protocol), minor visitor restrictions for ICU.

C1.2 ABAC Access Control

Role	Access	ABAC Conditions
Reception/Security (Visitor Desk)	V,C,U,X	VIEW: Patient bed/ward info (limited: name, ward, bed number only). CREATE: Visitor records. EXECUTE: Issue/revoke visitor passes. CANNOT: View clinical data. ABAC: Visitor module is isolated from clinical HMS.
Ward Nurse	V,A	VIEW: Visitor list for their ward. APPROVE: Special visit requests (outside hours, extra visitors). CANNOT: Access visitor personal data beyond name/pass number.
Treating Doctor	A	APPROVE: Visitor restrictions (no visitors for isolation patients), visitor exceptions. Execute via mobile app.
Infection Control	V,A,P	VIEW: Visitor traffic analytics. APPROVE: Outbreak mode visitor restrictions. PRINT: Visitor compliance reports.

C2. Patient Feedback & Grievance Redressal

C2.1 Functional Scope

Real-Time Feedback: Bedside tablet/kiosk feedback (emoji-based for simplicity). Post-discharge feedback via SMS/WhatsApp link (24 hours after discharge). NABH mandated domains: Doctor interaction, nursing care, cleanliness, food quality, billing transparency, overall satisfaction. NPS (Net Promoter Score) calculation.

Complaint Management: Multi-channel intake (kiosk, portal, WhatsApp, email, phone, in-person). Complaint categorization (clinical care, billing, staff behavior, infrastructure, waiting time, food, housekeeping). Auto-assignment to responsible department. SLA-based escalation: Level 1 (department): 24 hours. Level 2 (Quality Manager): 48 hours. Level 3 (Medical Superintendent): 72 hours. Level 4 (Grievance Committee): 7 days.

Grievance Committee: Meeting scheduling, agenda preparation from pending complaints, resolution documentation, patient notification of outcome, appeal process. NABH requires documented grievance redressal mechanism with patient awareness.

Analytics: Department-wise complaint trends, repeat complaints, resolution time trends, patient satisfaction scores over time, correlation with clinical outcomes.

C2.2 ABAC Access Control

Role	Access	ABAC Conditions
Quality Manager	V,C,U,A,P	FULL ACCESS to feedback system. VIEW: All complaints, feedback scores. APPROVE: Complaint closure, escalation. PRINT: Feedback reports for NABH. CREATE: Corrective action plans.
Department HOD	V,U,A	VIEW: Complaints assigned to their department. UPDATE: Response/action taken. APPROVE: Closure at department level. CANNOT: See other department complaints or patient-level feedback details for other areas.
Patient Relations Officer	V,C,U	VIEW: All active complaints. CREATE: New complaint registrations, follow-up notes. UPDATE: Status, patient communication records. CANNOT: Close complaints without Quality Manager approval.
Patient	V,C	VIEW: Own complaint status. CREATE: New complaint/feedback. CHANNEL: Portal, kiosk, WhatsApp. Verified by UHID or phone OTP.
Medical Superintendent	V,A,P	VIEW: All complaints (especially Level 3+). APPROVE: Final resolution for escalated complaints. PRINT: Monthly grievance summary.

C3. Queue Management System

C3.1 Functional Scope

OPD Token System: Patient checks in (kiosk/app/reception) → Token number generated → Token displayed on waiting area screen → SMS notification 3 patients before turn → Doctor calls next (from HMS screen) → Token displayed at doctor's room screen. Average waiting time displayed per doctor/department. Priority queue for: Disabled, elderly (> 70), pregnant, emergency referral.

Investigation Queue: Lab sample collection queue, radiology appointment queue, pharmacy dispensing queue. Estimated wait time calculation based on current processing rate.

Bed Admission Queue: When beds are full — waiting list management. Priority: Emergency > ICU step-down > Elective surgery > General admission. Waiting time tracking. Auto-notification when bed available.

Analytics: Real-time occupancy heatmap, average wait time per department/doctor, peak hours identification, bottleneck analysis, patient flow optimization.

C3.2 ABAC Access Control

Role	Access	ABAC Conditions
Queue Administrator	V,C,U,A,X	FULL queue management. APPROVE: Priority overrides. EXECUTE: Queue reset, display management. VIEW: Real-time analytics.
Receptionist	V,C,X	VIEW: Queue status for registration counters. CREATE: Token issuance. EXECUTE: Check-in, routing.
Doctor	V,X	VIEW: Own queue. EXECUTE: Call next patient, hold, skip. CANNOT: See or manage other doctors' queues.
Patient	V	VIEW: Own position in queue, estimated wait time. CHANNEL: App, SMS, display screen.

C4. Comprehensive Consent Management

C4.1 Functional Scope

This is one of the most legally critical modules. Every consent must be legally valid, properly documented, and readily retrievable.

Consent Types: General consent (admission, routine treatment), Surgical/procedure-specific consent (risks, benefits, alternatives in patient's language), Anesthesia consent, Blood transfusion consent, HIV testing consent, High-risk procedure consent (with specific risk percentages), Informed refusal (patient refuses recommended treatment — more important than consent), DAMA/LAMA consent, Photography/video consent (for documentation, telemedicine, teaching), Teaching/student observation consent, Research participation consent, ABDM/ABHA health data sharing consent, Organ donation consent, Advance directives (end-of-life wishes), DNR consent, DPDP Act data processing consent.

Multi-Language Generation: Consent forms auto-generated in patient's preferred language (English, Hindi, Tamil, Telugu, Kannada, Malayalam, etc.). Procedure-specific risk information in layman terms. Read-aloud option for illiterate patients (audio + witness signature).

Digital Signature: Aadhaar-based e-Sign (legally valid under IT Act). Biometric consent (thumbprint with witness). Digital pen signature on tablet. Video consent recording for high-risk procedures (patient stating understanding in own words).

Consent Verification: Before any procedure, system checks: Required consent(s) signed? Signed by appropriate person (patient, guardian for minor, NR for psychiatric)? Not expired (time-limited consents for specific procedures)? Not revoked? If consent missing: HARD BLOCK on procedure (only emergency override possible).

C4.2 ABAC Access Control

Role	Access	ABAC Conditions
Treating Doctor	V,C,U,A,X	VIEW: All consents for their patients. CREATE: Procedure-specific consent generation (system generates template, doctor customizes risks/alternatives). APPROVE: Consent adequacy. EXECUTE: Lock consent (no further modification). SPECIAL: Informed refusal documentation (more detailed than consent).
Nurse	V,C,X	VIEW: Consent status for assigned patients (signed/pending). CREATE: Witness documentation, general consent completion. EXECUTE: Verify consent before procedure (scan patient + consent barcode). CANNOT: Modify consent content.
Patient/Guardian	V,C	VIEW: All consents signed by them. CREATE: Revocation of consent (at any time). CHANNEL: Portal shows consent documents + explanatory videos. Can request clarification before signing.
Medical Records	V,P	VIEW: All consent documents. PRINT: Consent copies for patient request, legal proceedings. CANNOT: Modify consents. ARCHIVE: Consent documents follow record retention policy.
Legal Department	V,P	VIEW: Consent documents for legal cases. PRINT: Certified copies. CANNOT: Modify. AUDIT: Access logged and notified to treating doctor.

C5. Incident Reporting & Patient Safety (Sentinel Events)

C5.1 Functional Scope

Incident Types: Medication errors (wrong drug, wrong dose, wrong patient, wrong route, wrong time), fall incidents, needle stick injury, blood transfusion reactions, hospital-acquired infections, surgical site infections, wrong-site surgery, patient elopement/absconding, equipment malfunction during patient care, pressure ulcers (hospital-acquired), patient identification errors, near-miss events (caught before reaching patient), adverse drug reactions (ADR — CDSCO PvPI reporting), sentinel events (death, permanent harm, surgery on wrong site/patient).

Anonymous Reporting Option: Staff can report near-miss events anonymously to encourage reporting culture (Just Culture approach). Anonymous reports cannot be traced to reporter (technically enforced, not just policy). However, incidents involving actual patient harm MUST be attributed (for investigation).

Root Cause Analysis (RCA): For sentinel events, system guides through formal RCA: Timeline construction, contributing factor analysis, root cause identification, action plan (corrective + preventive), implementation tracking, effectiveness verification. RCA template per NABH standard.

CAPA Management: Corrective and Preventive Action tracking. Action assigned to responsible person. Due date. Follow-up. Effectiveness check (did the action prevent recurrence?). Closure with Quality Manager approval.

Regulatory Reporting: CDSCO Pharmacovigilance (ADR reporting via PvPI — Pharmacovigilance Programme of India), Hemovigilance (transfusion reactions to NACO), Materiovigilance (equipment incidents to CDSCO MvPI), NABH sentinel event reporting.

C5.2 ABAC Access Control

Role	Access	ABAC Conditions
Any Staff (Reporter)	C	CREATE: Incident reports (anonymous or attributed). Can report any incident regardless of department. No approval required to submit. Report immediately enters review queue. CANNOT: View other reports or analytics.
Department HOD	V,U,A	VIEW: Incidents in their department. UPDATE: Add investigation findings. APPROVE: Department-level action plan. CANNOT: View anonymous reporter identity. CANNOT: Delete or modify original incident report.
Quality Manager	V,C,U,A,P,X	FULL ACCESS to all incidents. CREATE: RCA documents, CAPA plans. APPROVE: Incident closure, CAPA effectiveness. EXECUTE: Regulatory report generation (PvPI, hemovigilance, materiovigilance). PRINT: NABH-format incident reports. SPECIAL: Only role that can see cross-department incident patterns.
Medical Superintendent	V,A,P	VIEW: All incidents (especially sentinel events). APPROVE: RCA findings, CAPA for serious incidents. PRINT: Monthly safety dashboard. NOTIFIED: Auto-alert for all sentinel events within 1 hour.
Patient Safety Officer	V,C,U,A,P	VIEW: All incidents. CREATE: Safety alerts, learning bulletins. APPROVE: Safety improvement initiatives. PRINT: Patient safety reports. EXECUTE: Hospital-wide safety alerts.

Critical ABAC Rules:

Rule INC-001: Anonymous reports: Reporter identity encrypted with a key held ONLY by the Quality Committee Chairperson. Even IT admin cannot decrypt. Legal protection under Quality Improvement and Patient Safety framework.

Rule INC-002: Incident reports are NEVER deletable. Even by system admin. They can be marked as duplicate or false alarm (with justification), but original remains in permanent audit log.

Rule INC-003: Medication error reports auto-notify: Prescribing doctor, dispensing pharmacist, administering nurse, department HOD, and Drugs & Therapeutics Committee chairperson.

C6. Hospital Committee Management

C6.1 Functional Scope

Mandatory Committees (NABH/NMC/Statutory): Hospital Infection Control Committee (HICC), Drugs & Therapeutics Committee (DTC), Quality Improvement Committee, Blood Transfusion Committee, Ethics Committee (IEC/IRB), Medical Audit Committee, Death Review Committee, Code Blue Committee, Disaster

Management Committee, Pharmacy & Therapeutic Committee, Anti-Ragging Committee, Internal Complaints Committee (Sexual Harassment — POSH Act), Grievance Redressal Committee, Condemnation Committee, Purchase Committee.

For Medical College additionally: Institutional Ethics Committee (ICMR guidelines), Medical Education Unit, Anti-Ragging Committee (UGC mandated), Internal Quality Assurance Cell (IQAC), Academic Council, Board of Studies per department, Research Advisory Committee, CBME Implementation Committee.

Meeting Management: Auto-scheduling (monthly/quarterly per committee charter). Agenda preparation (auto-populate from pending items — incidents, purchase requests, audit findings). Attendance tracking. Minutes recording. Action item generation with assignee + deadline. Follow-up tracking. Compliance reporting (NABH requires evidence of regular committee meetings with documented outcomes).

C6.2 ABAC Access Control

Role	Access	ABAC Conditions
Committee Chairperson	V,C,U,A,P	FULL ACCESS to own committee. APPROVE: Meeting minutes, action item closure. CREATE: Special meeting requests, emergency agenda items. PRINT: Committee reports. VIEW: Only own committee data.
Committee Secretary	V,C,U,P	CREATE: Agenda, minutes, action items. UPDATE: Action item status based on member feedback. PRINT: Meeting documents. CANNOT: Approve minutes (chairperson role).
Committee Member	V,C	VIEW: Meeting schedule, agenda, minutes. CREATE: Agenda suggestions, action item updates for items assigned to them. CANNOT: Access other committee data.
Quality Manager	V,P	VIEW: All committee meeting status (compliance dashboard — which committees met this month, which are overdue). PRINT: Committee compliance report for NABH. CANNOT: Access internal committee deliberations unless a member.

C7. Document Control & SOP Management

C7.1 Functional Scope

NABH and NABL both mandate a formal Document Control system. This is not just file storage — it's a controlled process.

Document Types: SOPs (Standard Operating Procedures), Policies, Clinical Guidelines/Protocols, Forms/Templates, Work Instructions, Quality Manual, Safety Manuals, Emergency Plans, Job Descriptions, Competency Assessment Tools.

Document Lifecycle: Draft (author creates) → Review (department experts review) → Approval (HOD/Quality Committee) → Release (controlled distribution) → Active (in use) → Periodic Review (annual) → Revision (if changes needed) → Obsolete (old version archived, watermarked 'OBSOLETE'). Version control: v1.0, v1.1, v2.0 etc. Every version preserved.

Controlled Distribution: Each released document has a distribution list. System tracks who has acknowledged receipt/reading of each document. Training requirement flagging (new SOP may require staff training — auto-creates training requirement in HR module).

Read Acknowledgment: When a new/revised SOP is released, affected staff get notification. Must read and acknowledge (digital signature) within defined timeframe (typically 7 days). Non-compliance auto-escalated to department HOD.

C7.2 ABAC Access Control

Role	Access	ABAC Conditions
Document Controller (Quality)	V,C,U,A,P,X	FULL ACCESS. CREATE: Document master records, distribution lists. APPROVE: Release of documents (after author + reviewer + approver chain). EXECUTE: Document release (triggers notifications), obsolescence. PRINT: Controlled copies (watermarked with copy number). VIEW:

		Complete document repository.
Author (Any Staff)	V,C,U	CREATE: Draft documents. UPDATE: Revisions when review cycle open. CANNOT: Release documents, modify after release. Each change tracked with change history.
Reviewer	V,U,A	VIEW: Documents sent for review. UPDATE: Add review comments. APPROVE: Reviewed status. CANNOT: Create, release, or modify final documents.
General Staff	V	VIEW: Released documents relevant to their department/role. Acknowledgment required for new/revised documents. CANNOT: Modify, print uncontrolled copies (printing creates watermarked 'UNCONTROLLED' copy).
External Auditor (NABH/NABL)	V,P	VIEW: Document master list, specific SOPs as needed for audit. PRINT: For audit purposes only. ACCESS: Time-limited, provisioned by Quality Manager.

C8. Hospital Communication Systems Integration

C8.1 Functional Scope

Code Systems: Code Blue (cardiac arrest), Code Red (fire), Code Pink (infant abduction), Code Orange (hazmat/chemical spill), Code Silver (weapon/hostage), Code Yellow (mass casualty), Code Black (bomb threat), Code White (aggressive patient/visitor), Code Brown (infrastructure failure — power, water, gas). Each code has: Activation trigger (manual/automatic), notification tree (who gets notified in what order), response team auto-assembly, documentation of response, deactivation authority, post-event debrief scheduling.

Nurse Call System Integration: Bedside call button → Nurse station panel → If unresponded in 2 min: Escalate to charge nurse → If unresponded in 5 min: Escalate to nursing supervisor. Call type: Routine, Urgent, Emergency. Response time tracking per call for nursing quality metrics.

Patient Communication: SMS gateway (appointment reminders, investigation ready, bill generated). WhatsApp Business API (report delivery, follow-up reminders, patient education materials). Push notifications (mobile app). Email (discharge summary, reports). IVR (automated appointment booking, report status check).

Clinical Communication: Critical value alerts (lab → doctor), clinical messaging (secure in-app messaging between clinical staff — NOT WhatsApp for clinical data), handover communication (SBAR format), referral communication, discharge communication to GP/referring doctor.

C8.2 ABAC Access Control

Role	Access	ABAC Conditions
Code Team Lead	V,C,A,X	EXECUTE: Code activation for assigned code type. APPROVE: Code deactivation. CREATE: Post-event documentation. VIEW: Response team availability.
IT/Communication Manager	V,C,U,A	VIEW: System status, message delivery metrics. CREATE: Communication templates. UPDATE: Notification configurations. APPROVE: SMS/WhatsApp template changes. CANNOT: View message content.
Clinical Staff	V,C,X	VIEW: Messages directed to them. CREATE: Clinical messages to specific recipients. EXECUTE: Acknowledge critical alerts. CANNOT: Access other staff messages or bulk communication.
Patient	V	VIEW: Messages sent to them. CANNOT: Initiate clinical communication (must go through portal/app request). CHANNEL: SMS, WhatsApp, Email, App notifications.

PART D: Medical College — Pre-Clinical & Administrative Modules

D1. Anatomy Department

D1.1 Functional Scope

Cadaver Management: This is highly regulated by NMC and state Anatomy Acts. Receipt: Source (unclaimed body from government hospital, donated body). Documentation: Police report/death certificate, magistrate order (for unclaimed), donor consent + family acknowledgment (for donated). Unique cadaver ID assignment.

Preservation: Embalming records (date, solution composition, volume injected, technician). Storage: Tank/slot allocation, formalin level monitoring, ventilation compliance. Condition monitoring: Fungal growth, deterioration assessment, usability period tracking.

Allocation & Utilization: Department-wise allocation (1st year MBBS batch size determines cadaver need — NMC norm: 1 cadaver per 10 students). Dissection hall scheduling. Prosected parts tracking. Usage per body region (head & neck, thorax, abdomen, upper limb, lower limb). Duration tracking (cadaver typically lasts one academic year).

Disposal: After complete utilization: Cremation/burial with dignity. Religious rites if applicable. Documentation of final disposal. Family notification if donated body. Annual inventory report to state Anatomy Act authority.

Museum: Specimen inventory (wet specimens in jars, dry specimens, plastinated specimens, models). Condition monitoring. Student/visitor access log. Specimen preparation records.

D1.2 ABAC Access Control

Role	Access	ABAC Conditions
Anatomy HOD/Professor	V,C,U,A,P,X	FULL ACCESS. CREATE: Cadaver receipt records, allocation orders. APPROVE: Disposal authorization, museum specimen addition. EXECUTE: Anatomy Act compliance reporting. PRINT: Inventory reports for NMC inspection.
Anatomy Faculty	V,C,U	VIEW: Cadaver allocation, dissection schedule. CREATE: Dissection session records, teaching documentation. UPDATE: Cadaver utilization status. CANNOT: Authorize receipt or disposal.
Anatomy Technician/Embalmer	V,C,U,X	VIEW: Cadaver inventory, preservation schedule. CREATE: Embalming records, preservation maintenance logs. EXECUTE: Formalin changes, tank maintenance. CANNOT: Access student data or academic records.
MBBS Student (1st Year)	V	VIEW: Dissection schedule, assigned cadaver/table, anatomy atlas. CANNOT: Access cadaver management records, preservation details, source information.

D2. Pre-Clinical & Para-Clinical Teaching Laboratories

D2.1 Common Lab Management

Lab Scheduling: Practical session scheduling (batch-wise), lab utilization tracking, faculty allocation per session, demonstrator assignment.

Reagent & Chemical Management: Inventory per lab (chemicals, reagents, stains, culture media). Procurement and stock tracking. MSDS (Material Safety Data Sheet) repository for each chemical. Expiry management. Consumption tracking per batch (cost per student calculation). Hazardous chemical register (state pollution board compliance).

Equipment Management (Teaching Labs): Each department has teaching-specific equipment separate from hospital equipment. Microscopes (bulk — 60-100 per batch), experimental kits, simulation models. Maintenance schedule, utilization per session, breakage/loss tracking.

Student Practical Records: Practical attendance, marks, experiments completed vs curriculum requirement. Practical exam scheduling, internal assessment.

D2.2 Department-Specific Notes

Physiology Lab: Hematology practicals (Hb estimation, blood grouping, TLC/DLC — student kits), clinical physiology equipment (ECG, spirometer, ergometer for teaching), amphibian/animal physiology equipment (if applicable — CPCSEA compliance required).

Biochemistry Lab: Student practical kits (qualitative analysis, quantitative estimation), semi-auto analyzers for teaching, chemical waste management (heavy metals, organic solvents).

Pharmacology Lab: Experimental pharmacology equipment (organ bath, polygraph/PowerLab — if animal experiments conducted). CPCSEA/IAEC compliance for animal experiments. Drug museum management. Prescription writing exercise documentation (CBME requirement). Animal House management if applicable: Animal procurement records, housing conditions, IAEC approval for each experiment, 3R principle compliance (Replace, Reduce, Refine).

Microbiology Lab (Teaching): Culture practice scheduling, biosafety training records per student, specimen handling competency, media preparation logs (separate from hospital microbiology).

Pathology Lab (Teaching): Slide library (thousands of teaching slides by system/disease), museum specimens (gross specimens), slide seminar scheduling, digital pathology for teaching, student microscopy competency assessment.

D2.3 ABAC Access Control (Common for all teaching labs)

Role	Access	ABAC Conditions
Department HOD	V,C,U,A,P	FULL ACCESS to department lab. APPROVE: Reagent/equipment procurement, chemical disposal, IAEC proposals. PRINT: NMC lab compliance reports.
Faculty/Demonstrator	V,C,U	VIEW: Lab schedule, student lists, equipment status. CREATE: Practical session records, student attendance/marks. UPDATE: Equipment condition reports. CANNOT: Approve procurement.
Lab Technician	V,C,U,X	VIEW: Schedule, inventory levels. CREATE: Reagent/chemical usage records, equipment maintenance logs. EXECUTE: Equipment setup for practicals, media preparation. CANNOT: Access student marks or academic records.
Student	V	VIEW: Lab schedule, practical manual, assigned experiments, own marks. CANNOT: Access lab management data, inventory, or equipment details.
IAEC Member (Pharmacology only)	V,A,P	VIEW: Animal experiment proposals, compliance records. APPROVE: Animal experiment protocols. PRINT: IAEC meeting minutes, annual report. ABAC: Only accessible when user.committee = IAEC.

D2.4 Equipment (Teaching Labs — Consolidated)

Equipment	Qty	Cost (INR)	Cal.	PM	Notes
Student Microscope (Binocular)	120	0.3-1L each	Annual	Annual	NMC norm: 1 per student in batch. Teaching attachment.
Hematology Kit (Student)	60	0.01-0.05L each	N/A	Per batch	Hb tube, lancets, glass slides, RBC pipettes
ECG Machine (Teaching)	2	0.5-1.5L each	Annual	Annual	Physiology practical: student records own ECG
Spirometer (Teaching)	2	0.3-1L each	Annual	Annual	PFT demonstration and student practice
Ergometer/Bicycle	1	0.5-2L	Annual	Annual	Exercise physiology practical
Semi-Auto Biochemistry Analyzer (Teaching)	2	3-8L each	Annual	6-Monthly	Student practicals, not clinical use

PowerLab/Polygraph (Pharmacology)	2	5-15L each	Annual	Annual	If animal experiments conducted. Computer-based data acquisition.
Organ Bath (Pharmacology)	10	0.1-0.3L each	Annual	Annual	Tissue pharmacology practicals
Biosafety Cabinet (Micro Teaching)	2	3-6L each	Annual	6-Monthly	Student biosafety training
Autoclave (Teaching Lab)	1/dept	1-3L each	Quarterly	Monthly	Media prep, decontamination
Microtome (Pathology Teaching)	2	2-5L each	Annual	6-Monthly	Student section cutting practice
Slide Scanner/Digital Pathology (Teaching)	1	10-30L	Annual	Annual	Virtual microscopy, remote teaching, exam

D3. Simulation Lab / Skills Lab / OSCE Center

D3.1 Functional Scope

This is increasingly mandatory for CBME implementation and NMC compliance. Many new medical colleges invest INR 1-5 crores in simulation.

Mannequin/Simulator Inventory: Low-fidelity (CPR mannequins, IV arm, suture pad), medium-fidelity (SimMan, SimMom — for scenario-based learning), high-fidelity (advanced patient simulators with programmable vital signs, voice, physiological responses). Each simulator tracked: Serial number, manufacturer, software version, maintenance history, usage hours, consumable tracking (skin pads, airway parts).

Scenario Library: Pre-built clinical scenarios mapped to CBME competencies. Scenario metadata: Competency addressed, difficulty level, clinical domain (medicine, surgery, OB-GYN, pediatrics, ER), learning objectives. Custom scenario creation by faculty.

OSCE Management: Station setup (typically 10-20 stations per exam). Station description (task, time, equipment, simulated patient brief). Student rotation scheduling (randomized). Examiner assignment. Score entry (real-time or paper-based). Standard setting (modified Angoff). Result analytics per station, per competency.

Session Recording & Debriefing: Video recording of simulation sessions (with student consent). Structured debriefing documentation (what went well, what to improve, action plan). Faculty debriefing competency tracking.

Competency Tracking: CBME requires demonstration of competencies at various levels (knows, knows how, shows how, does). Simulation sessions mapped to EPA (Entrustable Professional Activities). Student competency dashboard showing progression.

D3.2 ABAC Access Control

Role	Access	ABAC Conditions
Simulation Center Director	V,C,U,A,P,X	FULL ACCESS. CREATE: Scenarios, session schedules, assessment templates. APPROVE: OSCE results, competency sign-offs, equipment procurement. EXECUTE: Exam administration. PRINT: Competency reports for NMC.
Simulation Faculty/Instructor	V,C,U,X	VIEW: Scenarios, student assignments, session schedules. CREATE: Session records, debriefing notes, assessment scores. EXECUTE: Run simulation scenarios, control mannequin during session. CANNOT: Modify OSCE exam results after submission.
Simulation Technician	V,C,U,X	VIEW: Equipment status, session requirements. CREATE: Equipment maintenance records, consumable usage, setup completion logs. EXECUTE: Mannequin setup, software loading, AV system management. CANNOT: Access student scores, academic data.
Student	V	VIEW: Own session schedule, own competency progress, debriefing feedback for own sessions. CANNOT: Access other students' data, scenario details before session, assessment criteria.
Examiner (OSCE)	V,C,X	VIEW: Station description, marking rubric, student list (during exam). CREATE: Scores during exam (real-time entry). EXECUTE: Submit scores. CANNOT: Modify scores after submission, view other station scores. TIME: Access only during exam window.

D4. Research Management & Institutional Ethics Committee (IEC)

D4.1 Functional Scope

Research Project Management: Proposal submission portal (PI details, study design, methodology, budget, timeline). Review workflow: Technical review → Ethics review → Institutional approval → Funding approval. Progress tracking (milestones, enrolled subjects, SAEs). Completion and publication tracking.

Ethics Committee (IEC/IRB): ICMR Bioethics Guidelines 2017 compliance. Committee composition tracking (1/3 non-medical, 1/3 women, external members, legal expert, lay person). Meeting scheduling (minimum monthly if active proposals). Proposal review: Initial review, expedited review (minimal risk), continuing review

(annual), amendment review. Informed consent form review and approval. SAE (Serious Adverse Event) reporting and review. Protocol deviation tracking. Study monitoring/audit.

CTRI Integration: Clinical Trials Registry of India registration tracking. ClinicalTrials.gov cross-registration. Trial results reporting compliance.

Plagiarism Management: Integration with plagiarism detection (Turnitin/iThenticate/Ouriginal). Thesis/dissertation similarity index tracking. Publication similarity check before submission.

Publication Tracking: Faculty-wise publication database. Journal indexing verification. H-index, i10-index calculation. Publication count for NMC faculty promotion criteria. Predatory journal detection warning.

D4.2 ABAC Access Control

Role	Access	ABAC Conditions
IEC Chairperson	V,C,U,A,P	FULL ACCESS to IEC module. APPROVE: Final ethics approval, SAE review outcomes. CREATE: Meeting agendas, review decisions. PRINT: Ethics approval certificates. VIEW: All proposals across all departments.
IEC Member (Internal/External)	V,C,A	VIEW: Proposals assigned for review. CREATE: Review comments, opinions. APPROVE: Individual reviewer recommendation (approve/revise/reject). CANNOT: See other reviewers' comments before meeting (blind review). TIME: Access to proposal 7 days before meeting, revoked after decision.
IEC Secretary	V,C,U,P,X	VIEW: All proposals, meeting calendar. CREATE: Meeting scheduling, communication to applicants. UPDATE: Proposal status. EXECUTE: Certificate generation, communication dispatch. PRINT: Meeting minutes. CANNOT: Approve proposals.
Principal Investigator (PI)	V,C,U	VIEW: Own proposals and their review status. CREATE: New proposal submission, amendments, progress reports, SAE reports. UPDATE: Ongoing study data. CANNOT: View other researchers' proposals, IEC deliberations about own proposal.
Research Participant	V	VIEW: Own consent document, study information sheet. CHANNEL: Participant portal or printed copy. CANNOT: Access study-level data. SPECIAL: Can withdraw consent at any time (system immediately notifies PI and IEC).

D5. Central Library Management

D5.1 Functional Scope

Physical Library: OPAC (Online Public Access Catalog) — book search, availability, location. Circulation: Issue, return, renewal, reservation, overdue tracking, fine calculation. Accession register (new book entry). Book loss/damage tracking. Periodicals management (journal subscriptions, issue tracking, binding).

Digital Library: E-resource management (e-journals, e-books, databases). Access authentication (IP-based for campus, proxy/VPN for off-campus). Usage analytics (most accessed journals, per-department usage, peak hours). INFLIBNET/NLIST integration (for government-funded e-resources). Digital repository (institutional — theses, dissertations, faculty publications).

NMC Compliance: NMC mandates minimum books per student per subject. System tracks: Current stock vs NMC requirement, gap analysis per subject. Titles as per NMC recommended book list. Library spending as percentage of college budget. Seating capacity vs student strength.

Reading Room: Seat booking (online), occupancy tracking, study room/discussion room reservation. WiFi usage analytics within library premises.

D5.2 ABAC Access Control

Role	Access	ABAC Conditions

Chief Librarian	V,C,U,A,P,X	FULL ACCESS. APPROVE: Book procurement, subscription renewals, write-off of lost/damaged books. EXECUTE: OPAC management, vendor management. PRINT: NMC compliance reports, usage analytics.
Library Staff	V,C,U,X	VIEW: All catalog data, circulation status. CREATE: Accession entries, circulation transactions. UPDATE: Book condition, location. EXECUTE: Issue/return processing, fine collection. CANNOT: Approve procurement, access patron personal data beyond library use.
Student	V,C,X	VIEW: Catalog search, own issued books, due dates, fines. CREATE: Reservation requests, book recommendations. EXECUTE: Self-checkout (if kiosk available), online renewal. LIMIT: Max 4 books at a time (configurable per category). ABAC: Access to e-resources restricted to enrolled students (verified via academic ERP enrollment status).
Faculty	V,C,X	VIEW: Full catalog, extended access to e-resources. CREATE: Book purchase recommendations (auto-route to HOD for endorsement). EXECUTE: Extended borrowing (30 days vs 14 for students). LIMIT: Max 10 books. SPECIAL: Inter-library loan request.

D6. CME, FDP & Conference Management

D6.1 Functional Scope

CME (Continuing Medical Education): Event creation (topic, faculty, date, type — lecture/workshop/hands-on), registration management, attendance tracking, certificate generation (auto-generated with unique certificate number, QR verification). Credit hours tracking per doctor (State Medical Council CME requirements). CME calendar publication.

FDP (Faculty Development Program): NMC mandates Basic Course in Medical Education (BCME) and Advanced Course in Medical Education Technology (ACMET) for faculty. Track: Which faculty has completed basic FDP, which has completed advanced, validity (typically 3 years). Auto-alert when FDP renewal due. Session documentation for internal FDP programs.

Conference Management: Abstract submission portal, review workflow (blind review), acceptance notification, registration (early bird/regular/late), payment tracking, session scheduling (parallel tracks), presentation upload, feedback collection, post-conference proceedings generation.

NMC Faculty Requirements: Track per-faculty: Total teaching hours per week (NMC minimum), research publications, conference presentations, FDP completion, CME credits, community service documentation. All these feed into faculty appraisal and promotion criteria.

D6.2 ABAC Access Control

Role	Access	ABAC Conditions
Academic Coordinator/Dean	V,C,U,A,P	FULL ACCESS. APPROVE: CME/FDP calendar, faculty nominations. VIEW: All faculty CME/FDP compliance status. PRINT: NMC faculty compliance reports.
CME Organizer (Faculty)	V,C,U,X	CREATE: CME event, schedule, faculty invitations. EXECUTE: Attendance capture, certificate generation. UPDATE: Event details. VIEW: Registration list, attendance for own events only.
Faculty (Participant)	V,C	VIEW: Own CME/FDP history, credits, upcoming events. CREATE: Registration for events. CANNOT: Access other faculty records.
External Participant	V,C	VIEW: Public event details, registration form. CREATE: Registration. CANNOT: Access any internal data.

D7. Internship Management

D7.1 Functional Scope

NMC Mandatory Rotations: Internship is 12 months with mandatory postings: Medicine (2 months), Surgery (2 months), OB-GYN (2 months), Pediatrics (1 month), Orthopedics (1 month), Ophthalmology (15 days), ENT (15 days), Community Medicine (2 months including rural posting), Casualty/ER (1 month), Anesthesia (15 days), Elective (15 days). System enforces completion of all postings before completion certificate.

Intern Logbook: Separate from PG logbook. Track: Procedures performed/assisted (with case linkage to HMS if done in-hospital), cases seen, skills acquired. Auto-populated from HMS when intern is assigned to a department and documented in clinical records. Supervisor assessment per rotation (knowledge, skills, attitude, professionalism).

Attendance & Leave: Minimum attendance requirement per posting (NMC mandates). Leave management (casual leave, medical leave — limited). If leave exceeds allowance: Posting extension auto-calculated. If total excess leave > 30 days: Additional extension beyond 12 months.

Completion & Registration: On successful completion of all rotations + attendance + logbook sign-off: Completion certificate generated. Recommendation letter for provisional registration with State Medical Council. Entire intern record archived for verification (council/employer queries).

D7.2 ABAC Access Control

Role	Access	ABAC Conditions
Internship Coordinator	V,C,U,A,P,X	FULL ACCESS. CREATE: Rotation schedule, posting assignments. APPROVE: Posting completion sign-off, leave approvals, completion certificates. EXECUTE: Certificate generation. PRINT: NMC-format internship reports.
Department Supervisor (Per Rotation)	V,C,U,A	VIEW: Interns assigned to their department. CREATE: Assessment entries, feedback. APPROVE: Rotation completion sign-off. CANNOT: View intern data from other rotations.
Intern	V,C,U	VIEW: Own rotation schedule, logbook, attendance, assessments. CREATE: Logbook entries (procedures, cases — manual entries for external cases). UPDATE: Logbook annotations. CANNOT: View peer data, modify supervisor assessments.
Dean/Principal	V,A,P	VIEW: Internship compliance dashboard. APPROVE: Final completion certificates. PRINT: Batch-wise completion status for NMC.

D8. Hostel & Residential Management

D8.1 Functional Scope

Room Allocation: Room inventory (boys/girls hostels, PG hostel, intern hostel, married PG accommodation). Room allocation algorithm (year-wise, category-wise if applicable). Roommate assignment. Room change requests. Room condition documentation (entry/exit inventory).

Mess Management: Menu planning (weekly rotation), meal count per meal per hostel, food quality feedback system, mess fee collection, vendor/caterer performance tracking. Special diet requests (medical, religious, vegetarian preference). Kitchen safety and hygiene audit trail.

Hostel Attendance: Biometric/app-based attendance. Late entry tracking (after 10 PM). Absence reporting (must inform warden). Parent notification for prolonged absence.

Maintenance: Room maintenance requests (plumbing, electrical, furniture, pest control). Work order generation to facilities team. Resolution tracking. Common area maintenance (TV room, reading room, gym, laundry room).

Fee Management: Hostel fee structure, payment tracking, overdue alerts. Mess fee (fixed + variable model). Caution deposit management. Refund processing at departure.

D8.2 ABAC Access Control

Role	Access	ABAC Conditions
Chief Warden	V,C,U,A,P	FULL ACCESS across all hostels. APPROVE: Room allocation, disciplinary actions, late-night permissions. VIEW: All hostel dashboards. PRINT: Reports for administration/NMC.
Hostel Warden	V,C,U,A	VIEW: Own hostel data only. CREATE: Attendance records, incident reports. APPROVE: Room change requests, maintenance closure for own hostel. UPDATE: Student room assignments.
Student (Resident)	V,C	VIEW: Own room details, mess menu, fee status, maintenance request status. CREATE: Maintenance requests, leave/outing requests, mess feedback. CANNOT: View other students' data or access admin functions.
Mess Manager	V,C,U	VIEW: Meal counts, feedback, menu. CREATE: Procurement records, daily meal records. UPDATE: Menu changes (with warden approval). CANNOT: Access student personal data or hostel room information.
Parent/Guardian	V	VIEW: Student's hostel attendance (absent/present), fee payment status. CHANNEL: Portal with OTP verification. CANNOT: Access hostel operational data.

D9. Student Welfare & Anti-Ragging

D9.1 Functional Scope

Anti-Ragging System (UGC Mandated): Online complaint submission (anonymous option available, 24/7). Complaint classification: Verbal, physical, sexual, cyber, exclusion. Auto-routing to Anti-Ragging Committee + Dean + UGC portal (if severe). FIR generation assistance for criminal ragging. Monitoring: Anti-ragging undertaking from every student + parent (annual, digital). Anti-ragging squad patrol records. Fresher welcome event documentation.

Internal Complaints Committee (ICC/POSH Act): Sexual harassment complaint management. Confidential intake. Committee hearing scheduling. Inquiry documentation. Timeline compliance (90-day resolution mandate). Interim relief tracking. Final report and action. Appeal management.

Student Counseling Cell: Appointment booking with counselor. Session records (encrypted, therapist-eyes-only). Referral to psychiatry if needed. Outcome tracking. Stress screening programs (PHQ-9/GAD-7 batch-level screening with consent). Peer support program management.

Mentor-Mentee System: Faculty mentor allocation (typically 1:15-20 students). Meeting scheduling (minimum quarterly per NMC). Meeting record documentation. Student concern tracking. Mentor feedback on student progress.

Disability Support: Identified students with disabilities (RPWD Act categories). Accommodation plans (extra time in exams, accessible seating, assistive technology). Implementation tracking. Faculty awareness notifications.

D9.2 ABAC Access Control

Role	Access	ABAC Conditions
Anti-Ragging Committee Chair	V,C,U,A,P	FULL ACCESS to anti-ragging module. VIEW: All complaints (including anonymous complainant ID — sealed, viewable only by Chair + Dean in joint access). APPROVE: Inquiry conclusions, punishment recommendations. PRINT: Committee reports for UGC.
ICC Presiding Officer (POSH)	V,C,U,A,P	FULL ACCESS to ICC module. STRICT ISOLATION: ICC data completely separate from all other modules. Not visible even to Dean unless ICC PO explicitly shares. PRINT: Inquiry reports (limited distribution).
Student Counselor	V,C,U	VIEW: Only appointments and own session notes. CREATE: Session records (encrypted). CANNOT: Access student academic records, disciplinary records, or share session data without student consent. ABAC: data.type = counseling_session is maximally restricted — not accessible by any other role including Dean/NMC inspector.
Faculty Mentor	V,C	VIEW: Own mentees' basic academic progress (attendance, internal marks). CREATE: Mentoring session records. CANNOT: Access other mentors' records, disciplinary data, counseling data.
Student	V,C	VIEW: Own mentor details, counseling appointments, complaint status. CREATE: Complaints (anonymous or attributed), counseling appointment requests. CANNOT: See other students' complaints, committee proceedings.

Critical ABAC Rules:

Rule STU-001: Anti-ragging complainant identity: Encrypted with dual-key (Committee Chair + Dean). Cannot be revealed to accused during inquiry (constitutional protection). Technical enforcement: Even database admin cannot read.

Rule STU-002: Counseling records: ZERO visibility to anyone except the counselor and the student. Not part of student record. Not accessible during disciplinary proceedings. Not shared with parents. Destroyed 1 year after student graduates.

Rule STU-003: POSH/ICC data: Legally mandated confidentiality. Breach of confidentiality is itself an offense under POSH Act. System enforces: No copy, no print (except by PO for formal proceedings), no export, watermark on all views.

D10. Stipend Management & Service Bond Tracking

D10.1 Functional Scope

Stipend Processing: PG students, interns, and fellows receive monthly stipends. Track: Amount (varies by year — PG-1, PG-2, PG-3), attendance-based calculation (deductions for unauthorized absence), bank account details, monthly processing, tax deduction (if applicable for higher stipends), arrears management when government revises stipend.

Service Bond Management: Many state government medical college seats come with mandatory service bonds. Track: Student name, admission category (government/management/NRI), bond amount, bond period (typically 1-5 years), service commencement date, service location, completion tracking, bond release processing. If service not completed: Bond forfeiture calculation, legal notice generation, collection tracking.

Scholarship & Fee Waiver: Government scholarships (SC/ST/OBC/EWS), fee waiver tracking, scholarship disbursement, utilization certificates. Compliance with scholarship conditions (attendance, academic performance).

D10.2 ABAC Access Control

Role	Access	ABAC Conditions
Finance Officer	V,C,U,A,P,X	FULL ACCESS. EXECUTE: Stipend processing, scholarship disbursement, bond forfeiture initiation. APPROVE: Stipend deductions, bond release. PRINT: Payment reports, TDS certificates.
Academic Coordinator	V,C,A	VIEW: Student eligibility status, attendance data. CREATE: Eligibility recommendations. APPROVE: Bond release recommendations (academic completion verified).
Student/PG	V,C	VIEW: Own stipend details, bond status, scholarship status. CREATE: Bank detail updates, bond queries. CANNOT: View other students' financial data.
Parent/Guarantor	V	VIEW: Bond status, payment obligations. CHANNEL: Portal with verified identity.

D11. Alumni Management & Placement

D11.1 Functional Scope

Graduate Database: Automatic transfer from student database upon graduation. Track: Batch, degree, specialization, current position, workplace, contact details (voluntary updates). PG entrance exam performance tracking (institutional NEET-PG score trends). Higher education tracking (how many went to superspeciality, abroad, etc.).

Alumni Portal: Profile management (alumni update their own info), institutional news, event invitations, donation/contribution portal, mentorship program (senior alumni mentoring current students), job board (alumni can post opportunities).

Placement Statistics: Required for NMC/NAAC reporting. Track: Employment rate within 1 year of graduation, average compensation, geographic distribution, sector distribution (government/private/corporate/research).

D11.2 ABAC Access Control

Role	Access	ABAC Conditions
Alumni Coordinator	V,C,U,A,P	FULL ACCESS. CREATE: Events, batch records. APPROVE: Alumni profile verifications. PRINT:

		Placement statistics for NMC.
Alumni	V,C,U	VIEW: Alumni directory (limited per privacy settings), events, mentoring opportunities. CREATE: Own profile updates, mentorship offers, job postings. UPDATE: Own contact details.
Current Student	V	VIEW: Alumni directory (limited), mentorship opportunities, job postings. CANNOT: Contact alumni directly — must go through formal mentorship program.

D12. NMC NARF, NAAC & Accreditation Management

D12.1 Functional Scope

NMC NARF (National Assessment and Rating Framework): Self-assessment data collection from across all modules. Performance indicators: Student outcomes, faculty qualifications, research output, infrastructure adequacy, hospital clinical material, governance. Evidence compilation (auto-pull from HMS + Academic ERP). Gap analysis against NARF criteria. Action plan generation for gap closure.

NAAC Accreditation: If applicable — Criterion-wise data compilation (Curricular Aspects, Teaching-Learning, Research, Infrastructure, Student Support, Governance, Institutional Values). SSR (Self-Study Report) data aggregation. DVV (Data Verification and Validation) evidence management. Peer team visit coordination.

NMC Inspection Readiness: Continuous dashboard showing: Current compliance status per NMC MSR parameter. Red/Yellow/Green indicators. Documents/evidence auto-compiled. Deficiency alerts (e.g., faculty vacancy, bed occupancy below minimum). Mock inspection scheduling with internal assessors.

D12.2 ABAC Access Control

Role	Access	ABAC Conditions
IQAC Coordinator	V,C,U,A,P,X	FULL ACCESS. CREATE: Self-assessment reports, action plans, compliance documentation. APPROVE: Data accuracy verification from departments. EXECUTE: Report compilation, evidence packaging. PRINT: NARF/NAAC reports.
Department HOD	V,C,U	VIEW: Own department compliance indicators. CREATE: Department-level evidence/data. UPDATE: Action plan progress for own department.
Dean/Principal	V,A,P	VIEW: Institutional-level compliance dashboard. APPROVE: Final NARF submission, NAAC SSR. PRINT: Executive summary reports.
NMC/NAAC Assessor (External)	V,P	VIEW: Pre-compiled assessment data package. PRINT: Required reports. ACCESS: Time-limited, read-only, fully audited. Provisioned by Dean.

PART E: Appendices

E1. Complete Module Inventory (All Modules Across v2.0 + v2.1)

#	Module	Category	Priority	Phase
1	Patient Registration + ABHA	Clinical	Must-Have	Phase 1
2	OPD (16 Specialties)	Clinical	Must-Have	Phase 1
3	IPD (Admission/Rounds/Discharge)	Clinical	Must-Have	Phase 1
4	Emergency Department	Clinical	Must-Have	Phase 2
5	ICU / HDU / CCU	Clinical	Must-Have	Phase 2
6	NICU / PICU	Clinical	Must-Have	Phase 2
7	Operation Theatre	Clinical	Must-Have	Phase 2
8	Laboratory (LIS)	Clinical	Must-Have	Phase 2
9	Radiology (RIS/PACS)	Clinical	Must-Have	Phase 3
10	Pharmacy	Clinical	Must-Have	Phase 1
11	Blood Bank	Clinical	Must-Have	Phase 3
12	Nursing Documentation	Clinical	Must-Have	Phase 2
13	Billing & Finance	Admin	Must-Have	Phase 1
14	HR & Payroll + AEBAS	Admin	Must-Have	Phase 2
15	Materials & Inventory	Admin	Must-Have	Phase 2
16	CSSD	Clinical	Must-Have	Phase 2
17	Dietary	Support	Should-Have	Phase 3
18	Infection Control	Clinical	Must-Have	Phase 3
19	Quality/NABH/NMC	Compliance	Must-Have	Phase 3
20	Academic ERP	Academic	Must-Have	Phase 3
21	Medico-Legal	Legal	Must-Have	Phase 2
22	Telemedicine	Clinical	Should-Have	Phase 4
23	Ambulance/Fleet	Support	Nice-to-Have	Phase 4
24	Equipment Management (MEMS/CMMS)	Infrastructure	Must-Have	Phase 1-2
25	ABAC Access Control Engine	Security	Must-Have	Phase 1
26	Break-Glass Emergency Access	Security	Must-Have	Phase 1
	—— NEW IN v2.1 ——			
27	Mortuary & Post-Mortem	Clinical	Must-Have	Phase 2
28	Cath Lab / Interventional	Clinical	Must-Have	Phase 3
29	Endoscopy Suite	Clinical	Must-Have	Phase 3
30	Day Care / Short Stay	Clinical	Must-Have	Phase 2
31	Palliative Care + DNR + Organ	Clinical	Should-Have	Phase 3

	Donation			
32	Nuclear Medicine	Clinical	Nice-to-Have	Phase 4
33	PMR (Physio+OT+Speech+Prosthetics)	Clinical	Should-Have	Phase 3
34	Audiology & Speech Pathology	Clinical	Should-Have	Phase 3
35	Psychiatry MHCA 2017 Compliance	Clinical/Legal	Must-Have	Phase 2
36	Clinical Psychology	Clinical	Should-Have	Phase 3
37	Medical Gas (MGPS) Monitoring	Infrastructure	Must-Have	Phase 2
38	Biomedical Waste Management	Infrastructure/Legal	Must-Have	Phase 2
39	Fire Safety & Emergency Prep	Infrastructure	Must-Have	Phase 2
40	Water Quality & Energy Management	Infrastructure	Should-Have	Phase 3
41	Laundry Management	Support	Should-Have	Phase 3
42	Security & Physical Access Control	Infrastructure	Should-Have	Phase 3
43	Visitor Management	Operations	Should-Have	Phase 3
44	Patient Feedback & Grievance	Operations	Must-Have	Phase 2
45	Queue Management	Operations	Must-Have	Phase 2
46	Consent Management (Comprehensive)	Legal/Clinical	Must-Have	Phase 1-2
47	Incident Reporting & Patient Safety	Quality/Safety	Must-Have	Phase 2
48	Hospital Committee Management	Admin	Must-Have	Phase 3
49	Document Control / SOP Management	Quality	Must-Have	Phase 2
50	Communication Systems Integration	Operations	Should-Have	Phase 3
51	Anatomy Department	Academic	Must-Have	Phase 3
52	Teaching Labs (Physio/Biochem/Pharma/Micro/Path)	Academic	Must-Have	Phase 3
53	Simulation Lab / OSCE Center	Academic	Must-Have	Phase 3
54	Research & Ethics Committee (IEC)	Academic	Must-Have	Phase 3
55	Central Library	Academic	Must-Have	Phase 3
56	CME / FDP / Conference Management	Academic	Should-Have	Phase 4
57	Internship Management	Academic	Must-Have	Phase 3
58	Hostel & Residential Management	Academic	Should-Have	Phase 4
59	Student Welfare & Anti-Ragging	Academic/Legal	Must-Have	Phase 3
60	Stipend & Service Bond Tracking	Academic/Finance	Must-Have	Phase 3
61	Alumni Management & Placement	Academic	Nice-to-Have	Phase 4
62	NMC NARF / NAAC Accreditation	Compliance	Must-Have	Phase 3
63	Forensic Medicine / Autopsy	Academic/Clinical	Must-Have	Phase 3
64	Parent/Guardian Portal	Academic	Should-Have	Phase 4
65	Follow-up & Defaulter Tracking	Clinical	Should-Have	Phase 3
66	Discharge Planning	Clinical	Should-Have	Phase 2
67	Patient Transport (Intra-hospital)	Operations	Nice-to-Have	Phase 4

Total modules: 67. Must-Have: 42, Should-Have: 18, Nice-to-Have: 7.

This is the most comprehensive HMS + Medical College ERP specification ever created for the Indian healthcare market. No competitor covers even 40% of these modules in a single integrated system.

E2. Updated Pricing Strategy

With 67 modules (vs. typical HMS offering 15-20 modules), our pricing strategy positions us as follows:

Tier	Our Price (INR)	Market Equivalent	Savings	Modules Included
NMC Medical College (300-500 beds)	35-40 Lakhs	1.2-1.5 Crore (4-5 vendors)	65-75%	All 67 modules. Unified. Single vendor.
NABH Hospital (200-500 beds)	20-28 Lakhs	50-80 Lakhs (3-4 vendors)	55-65%	45+ modules (hospital modules + CMMS + compliance)
Multi-Specialty (100-200 beds)	12-18 Lakhs	25-40 Lakhs	50-55%	35+ modules (core clinical + admin + CMMS)
Small Hospital (< 100 beds)	5-10 Lakhs	12-25 Lakhs	45-55%	25+ modules (essential clinical + admin)
Standalone Lab (NABL)	3-5 Lakhs	8-15 Lakhs	50-60%	LIS + Lab CMMS + NABL compliance + QC
Clinic/Polyclinic	1-3 Lakhs	3-8 Lakhs	55-65%	OPD + Pharmacy + Billing + Basic CMMS

The pricing premium of INR 5L over v2.0 estimate is justified by: 43 additional modules, comprehensive ABAC engine (which alone would cost INR 5-10L if custom-built), integrated CMMS (market price INR 10-20L), medical college modules (no competitor offers this at any price), and legal compliance modules (MHCA, PCPNDT, BMW Rules, AERB — each requiring domain expertise that consultants charge INR 2-5L to implement).